

Industrial Internship Report on

"URL Shortener"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 4 weeks' time.

The project assigned during the internship was URL Shortener, a web-based application developed using Python, Flask, SQLite, HTML, and CSS. The primary objective of the project was to simplify long web addresses by converting them into short, unique, and easy-to-share links. Whenever a user accesses the generated short URL, the application retrieves the corresponding original URL from the database and redirects the user to the intended website.

The project was completed by following a structured software development process that included requirement analysis, system design, implementation, testing, debugging, documentation, and deployment preparation. Throughout the internship, various concepts such as Flask routing, database management with SQLite, HTML form handling, URL validation, and GitHub version control were learned and applied successfully.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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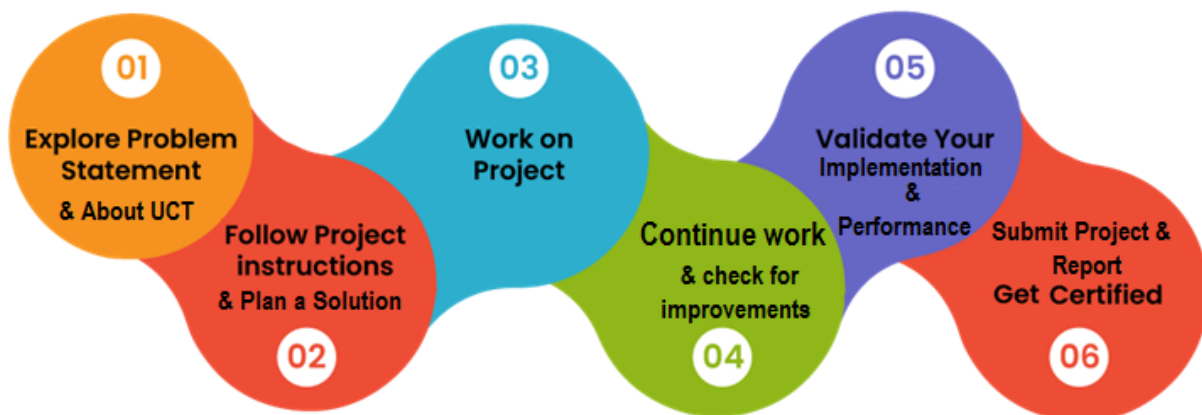
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1 Preface

Industrial internships play an important role in bridging the gap between classroom learning and practical industry experience. They provide students with an opportunity to apply theoretical concepts to real-world problems while developing technical and professional skills. This internship, organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT), provided valuable exposure to software development using Python and web technologies.

During the six-week internship, I worked on the URL Shortener project. The objective of this project was to develop a web application capable of converting long URLs into short, unique, and easily shareable links. The application stores the mapping between the original URL and the generated short URL in a SQLite database. Whenever a user accesses the shortened link, the application retrieves the corresponding original URL from the database and redirects the user automatically.

The internship followed a systematic approach throughout the development process. Initially, the project requirements were analyzed, followed by designing the application architecture and database structure. The project was then implemented using Python, Flask, HTML, CSS, and SQLite. After implementation, extensive testing and debugging were performed to ensure that all functionalities worked correctly. Finally, the project was documented, organized, and prepared for GitHub submission.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



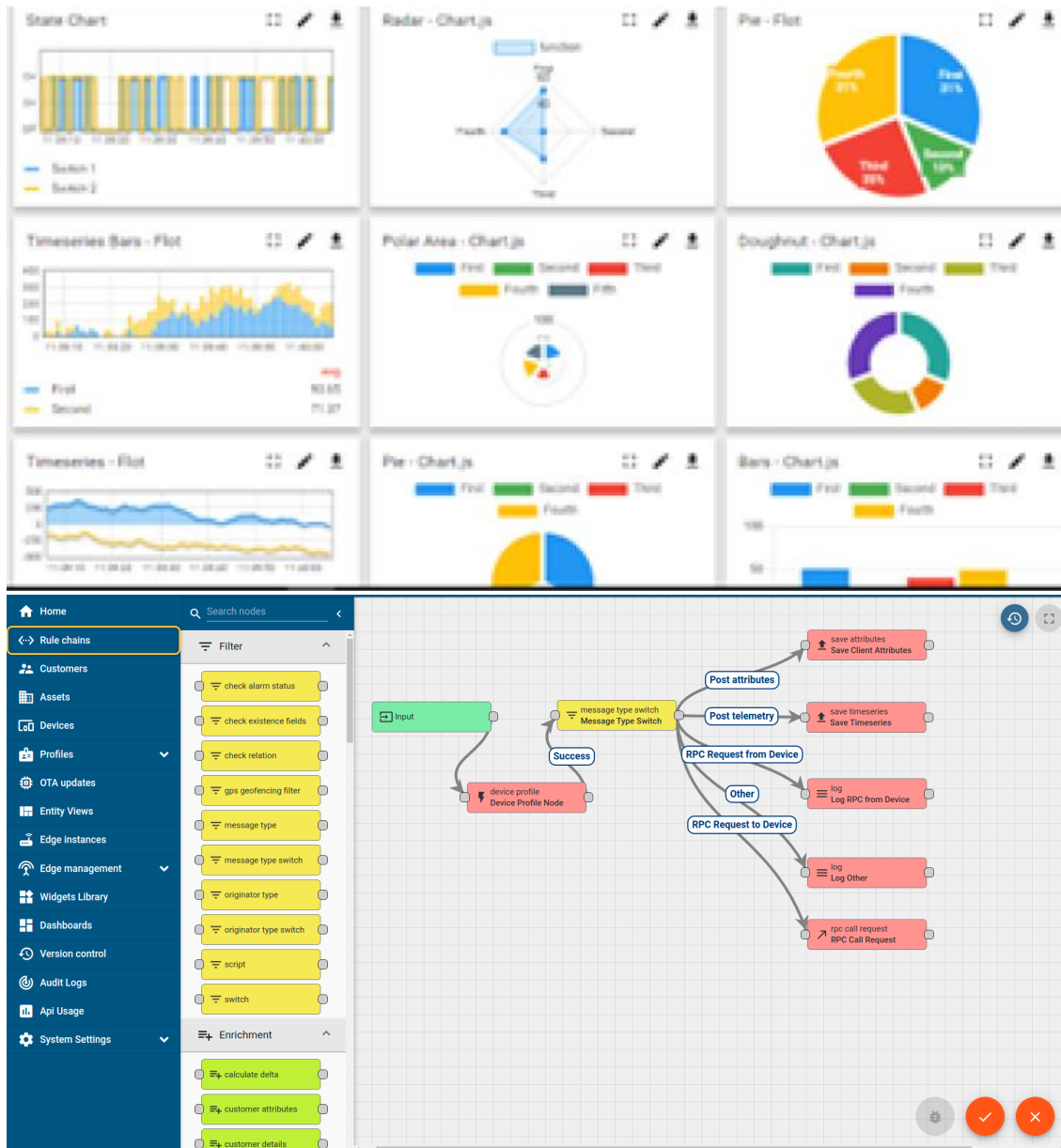
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

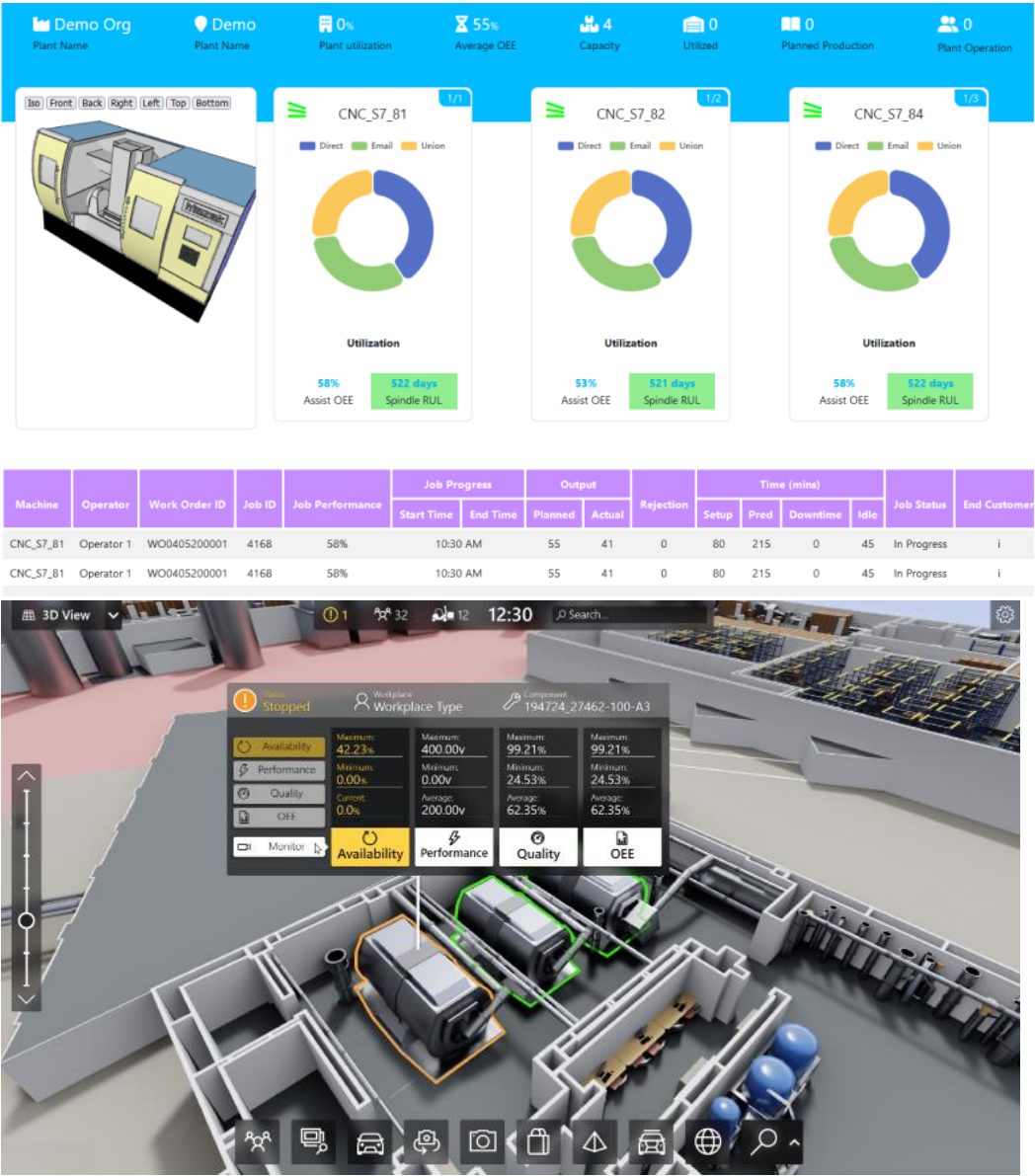
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



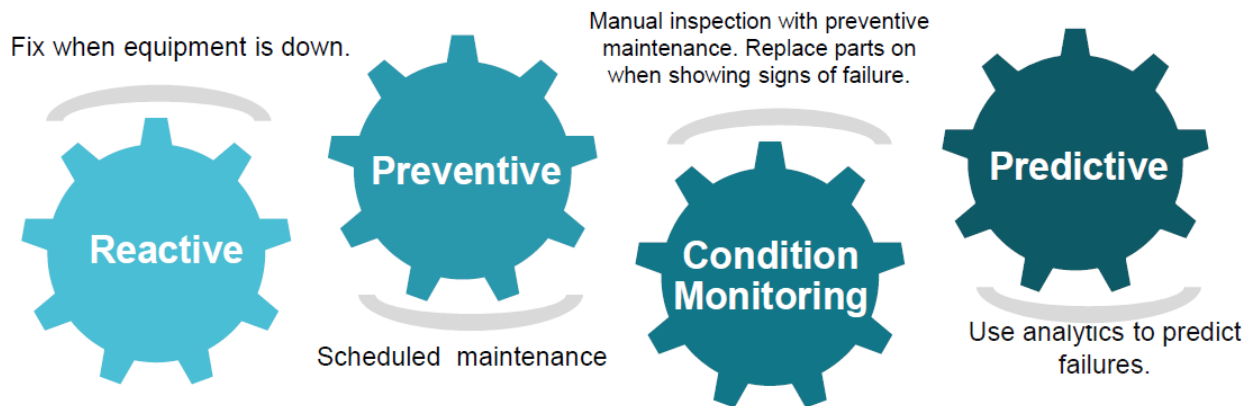


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

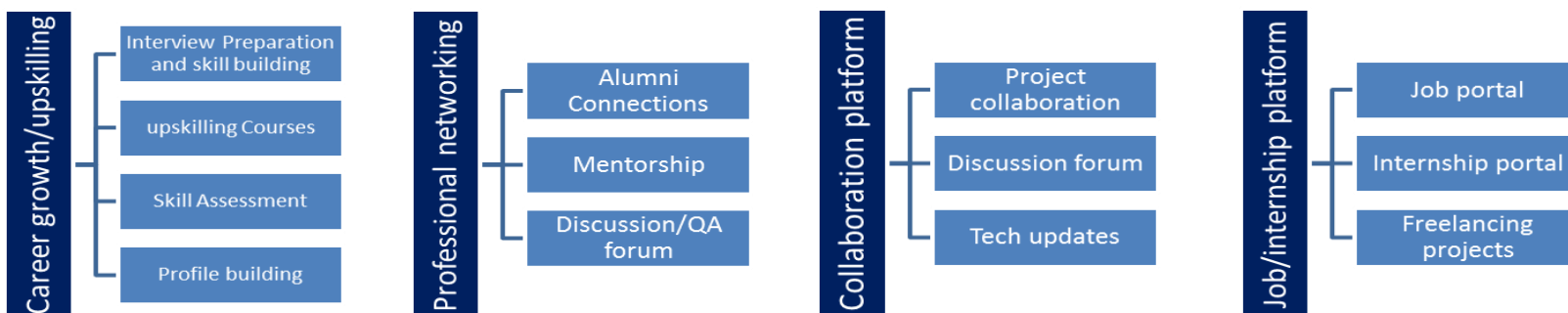
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python and Flask official documentation
- [2] HTML and CSS documentation
- [3] Github documentation

2.6 Glossary

Terms	Acronym
URL	Uniform Resource Locator used to identify web resources.
Flask	A lightweight Python web framework used for building web applications.
SQLite	A lightweight relational database management system.
HTML	HyperText Markup Language used to create web pages.
CSS	Cascading Style Sheets used for designing web pages.

3 Problem Statement

During the development of this project, the following functional requirements were identified:

- Accept a valid long URL from the user.
- Generate a unique short code for each URL.
- Store the original URL and short code in a SQLite database.
- Redirect users from the short URL to the original URL.
- Display the generated short URL on the web interface.
- Prevent invalid or empty URL submissions.
- Ensure reliable storage and retrieval of URL information.

Challenges Identified

Several challenges were considered while developing the project:

- Generating unique short codes without duplication.
- Validating user input before storing URLs.
- Efficiently managing URL mappings within the database.
- Handling invalid or expired short URLs gracefully.
- Providing a simple and responsive user interface.

Expected Outcome

The expected outcome of this project is a lightweight web application capable of generating short URLs quickly and accurately. The application should simplify URL sharing, improve user convenience, and demonstrate the practical implementation of Python web development concepts such as routing, database management, form handling, and URL redirection.

4 Existing and Proposed solution

Existing Solution

Many online platforms such as TinyURL and Bitly provide URL shortening services. These platforms convert long URLs into short links that are easier to share. However, they are hosted online, may require user registration for advanced features, and do not provide insight into the implementation for learning purposes.

Limitations of Existing Solutions

- Dependence on third-party services.
- Limited customization in free versions.
- Internet connection required.
- Not suitable for understanding the internal working of URL shortening.

Proposed Solution

The proposed solution is a URL Shortener developed using Python, Flask, SQLite, HTML, and CSS. The application generates a unique short URL for every valid long URL, stores the mapping in a SQLite database, and redirects users to the original website when the short URL is accessed.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

5.1 System Architecture

The URL Shortener is designed using a straightforward client-server architecture. A user submits a long URL through the web interface, and the Flask application processes the request by generating a unique short code, storing the URL mapping in the SQLite database, and returning the shortened URL. When the shortened URL is accessed, the application retrieves the corresponding original URL from the database and automatically redirects the user.

5.2 Database Design

The project uses a SQLite database with the following table structure:

Field	Data Type	Description
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short	TEXT	Short URL code
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original	TEXT	Original long URL
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clicks	INTEGER	Number of times the short URL is accessed
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5.3 Workflow

1. The user enters a long URL.
2. Flask validates the URL.
3. A unique short code is generated.
4. The URL mapping is stored in the SQLite database.
5. The generated short URL is displayed to the user.
6. When the short URL is accessed, the application retrieves the original URL and redirects the user.

6 Performance Test

6.1 Test Plan/ Test Cases

The application was tested by entering different valid and invalid URLs through the web interface. The generated short URLs were opened to verify proper redirection. Additional tests were performed to check database storage, click tracking, and input validation.

6.2 Test Procedure

The application performed successfully during testing. All core functionalities, including URL generation, database storage, redirection, and validation, worked as expected. The system was stable, responsive, and suitable for small-scale web application usage.

6.3 Performance Outcome

The URL Shortener demonstrated reliable performance throughout testing. Short URLs were generated quickly, database operations were completed efficiently, and users were redirected to the original websites without noticeable delay. Overall, the application met the functional requirements and achieved the expected project objectives.

7 My learnings

The internship provided me with practical exposure to software development using Python and modern web technologies. It helped me understand how theoretical concepts learned in the classroom can be applied to develop real-world applications.

During the development of the URL Shortener project, I gained hands-on experience with the Flask framework for creating web applications and SQLite for database management. I learned how to design application workflows, handle user requests, generate unique short URLs, and implement URL redirection efficiently.

I also improved my knowledge of HTML and CSS while designing a simple and user-friendly interface. Through continuous testing and debugging, I developed better problem-solving skills and learned the importance of writing clean, maintainable, and efficient code.

In addition, I became familiar with GitHub for version control and project management. Preparing project documentation and reports enhanced my technical writing and presentation skills.

Overall, this internship strengthened my programming knowledge, increased my confidence in developing web applications, and provided valuable industry-oriented experience that will be beneficial for my future academic projects and professional career.

8. Future Work Scope and Conclusion

8.1 Future Work Scope

Although the current URL Shortener application successfully meets its objectives, there are several opportunities for future improvements. Additional features can enhance the functionality, security, and overall user experience of the application.

Some possible future enhancements include:

- User registration and login functionality.
- Custom domain support for shortened URLs.
- QR code generation for every short URL.
- Advanced analytics to monitor clicks and user activity.
- URL expiration and automatic deletion of inactive links.

8.2 Conclusion

The URL Shortener project was successfully designed and implemented using Python, Flask, SQLite, HTML, and CSS. The application provides a simple and efficient solution for converting long URLs into short, easy-to-share links while maintaining reliable redirection to the original websites.

Throughout the internship, the project was developed by following a systematic software development process that included requirement analysis, system design, implementation, testing, debugging, and documentation. Each stage contributed to improving both the quality of the application and my understanding of software engineering practices.

The project enhanced my knowledge of Python programming, Flask web development, database management, frontend development, and version control using GitHub. It also strengthened my analytical thinking, debugging skills, and ability to solve practical programming problems.

Overall, the internship was a valuable learning experience that provided practical exposure to real-world software development. The knowledge and skills gained during this project will be beneficial for future academic work as well as my professional career in software development.

